

Week 36, Tuesday Session

Flipped classroom: your questions, then bias-variance and cross-validation in practice

Bring your laptop – we work in `week36tuesday.ipynb`

Morten Hjorth-Jensen

Department of Physics and Center for Computing in Science Education,
University of Oslo, Norway

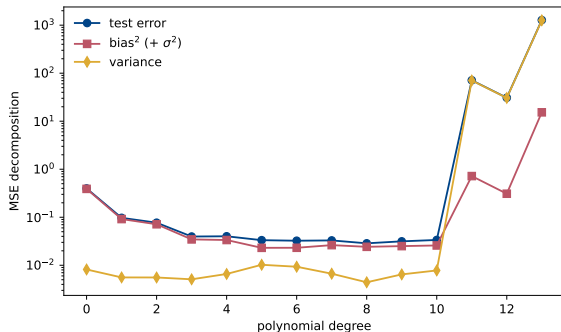
How today works

The format (as every Tuesday)

Mondays are lectures; Tuesdays belong to your questions and to concrete cases. You have heard the lecture and skimmed Sections 2.3–2.12 and 3.14.

1. **First ~20 min:** your open questions from Monday's lecture and from the reading, discussed in plenum.
2. **Rest of the double session:** *two* cases only this week – the bias-variance decomposition, and cross-validation done properly, for Ridge and for the Lasso – worked in pairs or small groups from `week36tuesday.ipynb`. Deep beats broad: the goal is that you leave owning Eq. (2.52) and the complete selection recipe of Section 3.14 – grid, folds, standard error, one-SE rule, refit, test set once – both of which Project 1 asks for.
3. **Last 5 min:** wrap-up, exercise deadline, and what next Monday builds on.

Picking up from Monday: the measured decomposition



Monday's lecture closed the loop on last week's U-curve: the test error is, by Eq. (2.52), the sum

$$\text{Bias}^2 + \text{Var} + \sigma^2,$$

and the bootstrap let us *measure* each term. Bias falls and flattens, variance explodes, the sum has a broad minimum (degrees 5–9 here). Today you reproduce this figure

yourselves, stress-test it – more data, more noise, a Ridge penalty – and then let cross-validation find the minimum without ever touching the test set, following the full recipe of Section 3.14 for Ridge and for the Lasso.

Case 1: the bias-variance decomposition

Goal: reproduce Monday's figure, then break it and understand the pieces (notebook, Case 1; seed 2026 gives exactly the lecture numbers).

1. Run the sweep ($n = 40, 100$ bootstraps, degrees 0–13) and reproduce `week36_bias_variance.pdf`.
2. **Vary the data size:** $n = 100$ and $n = 400$. Where does the variance explosion move? Does the bias curve care?
3. **Vary the noise:** $\sigma = 0.3$. Predict first, then check: what happens to the floor and to the minimum?
4. **Swap the model:** replace OLS by Ridge with $\lambda \in \{10^{-4}, 10^{-1}, 10\}$ at fixed high degree 12. Watch the variance fall and the bias rise as λ grows – the tradeoff turned into a dial.

Checkpoints (plenary after ~ 25 min)

Degree 0 has $\text{bias}^2 \approx 0.39$ ($\approx$ the variance of f itself – why?); the minimum is broad, degrees 5–9; at degree 13 the variance is $\sim 10^3$. For step 4: which λ gives the smallest *total* error at degree 12, and is it better than plain OLS at degree 6?

Case 2: cross-validation: choosing λ without touching the test set

Goal: a k -fold loop using `scikit-learn` or your own code, then used for real model selection (notebook, Case 2, steps 1–4).

1. Implement 5-fold CV for Ridge on $y = 3x^2 + \varepsilon$ (degree-6 polynomial, λ on `np.logspace(-3, 5, 100)`), scaler fitted *inside* each fold. Reproduce `week36_cv_ridge.pdf`.
2. Read off λ^* and compare the minimal CV-MSE with the known $\sigma^2 = 1$. Vary $k \in \{5, 10, 20\}$: how stable are λ^* and the error bar (the spread over folds)?
3. **Close the week's loop:** run CV over the polynomial *degree* for the week-35 data set (OLS, degrees 0–13) and compare the CV curve with Case 1's test-error curve – same minimum, no test set consumed.

Checkpoints

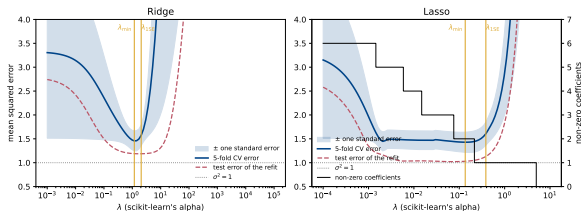
With seed 2026: $\lambda^* \approx 1.2$, CV-MSE ≈ 1.5 (just above the noise floor). For step 3: CV picks a degree in the same broad 5–9 window as the bootstrap decomposition, that is two independent methods, one answer.

Case 2, continued: the full recipe of Section 3.14 – Ridge *and* Lasso

Goal: turn the loop into the complete, honest procedure (notebook, steps 5–8).

- 4. Standard error and the one-SE rule.** Keep the spread over the folds, Eq. (3.93). Find $\lambda_{\min}$ and the largest λ within one standard error of it, Eq. (3.96). **Refit** the pipeline at each on *all* 100 training points and evaluate **once** on a fresh test set of 1000 points.
- 5. The Lasso, same data, same folds.** Where are $\lambda_{\min}$ and λ_{1SE} , how many of the six powers survive, and which? (Mind the conventions: Ridge's α is the book's $n\lambda$, Lasso's is $\lambda/2$ – Sections 3.10 and 3.14.)
- 6. Bootstrap versus cross-validation.** Compute the training error, the leave-one-out bootstrap, Eq. (3.98), and the .632 estimator, Eq. (3.99), for Ridge at $\lambda \in \{10^{-3}, \lambda_{\min}, \lambda_{1SE}, 10^3\}$ and set them beside CV and the test error. Which are optimistic, which pessimistic, and where does each put its minimum?
- 7. Which coefficients do you trust?** Refit the Lasso at $\lambda_{\min}$ and λ_{1SE} to 200 bootstrap replicas and count how often each power is selected.

Case 2, continued: what you should find



Checkpoints (seed 2026)

- ▶ Ridge: $\lambda_{\min} \approx 1.2$ (CV 1.46 ± 0.21), $\lambda_{1SE} \approx 2.1$; both refits test at 1.19 – the folds cannot tell them apart, and neither can the test set.
- ▶ Lasso: $\lambda_{\min} \approx 0.14$ keeps x^2 and a small x^4 (test 1.03, on the floor); $\lambda_{1SE} \approx 0.39$ keeps x^2 *alone* – the data were $3x^2 + \varepsilon$.
- ▶ Bootstrap: LOO-bootstrap 2.7 and .632 2.2 against a test error of 1.19 at $\lambda_{\min}$ – bootstrap models see only 63% of the points; CV (1.46) is closer. All put the minimum in the same flat valley.
- ▶ Stability: x^2 selected in 100% of replicas, x^4 in 58% (34% at one SE) – CV chose λ , the bootstrap says which coefficient to believe.

Plenary question: why is the CV estimate (1.46) *above* the test error (1.19) here, and why is that the safe direction to be wrong in?

Wrap-up, deadlines, and Monday

- ▶ **What you should own now:** Eq. (2.52) and what each term means operationally; a bootstrap loop; the selection recipe of Section 3.14 (leak-free folds, standard error, one-SE rule, refit, test set once) for Ridge and Lasso; and one sentence on why cross-validation, not the bootstrap, chooses λ .
- ▶ **Weekly exercises** (`exercisesweek36.ipynb`): the analytical steps of the bias-variance derivation, plus bootstrap and cross-validation on your own code. Deadline **Friday September 4 at midnight**.
- ▶ **Project 1** asks for exactly today's machinery – the code you wrote today is directly reusable.
- ▶ **Next Monday and week 38:** the main focus shifts to *optimization*: gradient descent methods, from plain and steepest descent via momentum to stochastic gradient descent, the workhorse behind everything from Lasso to deep networks. We will also briefly mention the Bayesian reading of OLS, Ridge and Lasso (a Gaussian prior gives Ridge, a Laplace prior the Lasso).

Before you leave

One line in an email to us: the muddiest point of the week – what should Monday's first ten minutes revisit?